Lecture 1. Introduction

Survival Data, Censoring, and Truncation

1 Course Overview

- Textbook: Klein & Moeschberger, *Survival Analysis: Techniques for Censored and Truncated Data*, 2nd ed. (cited below as K&M). These notes follow its Chapters 2–5, 7–9, 11 and 12.
- The emphasis is on methodology and application, with part of the underlying theory (why the methods work and how they were developed).
- Prerequisites: basic probability theory, statistical inference, and R programming.

2 What Is Survival Data?

Definition 1.1: Survival data

Survival data track the *time until an event of interest occurs*:

$$\text{Exposure (clock starts)} \xrightarrow{\text{survival time}} \text{Event (clock ends)}.$$

Examples: time from diagnosis to death; time from treatment to cancer recurrence; time to failure of a machine or component; time to staff promotion; time to divorce; time to unsubscription.

Example 1.2: Worcester Heart Attack Study (WHAS100)

Survival times following hospitalisation for a heart attack (MI). A snapshot:

ID	Female	Age (years)	Follow-up (years)	Event (death)
16	0	39	7.2	0
66	1	91	1.0	1
71	0	65	5.1	0
55	0	80	5.3	0
61	0	90	0.3	1

Question: what is the mean survival time after hospitalisation?

- *Wrong approach 1* (use only subjects with known survival times): $(1.0 + 0.3)/2 = 0.65$. This is an **underestimate**, because we ignored the people who survived longer.
- *Wrong approach 2* (pretend follow-up time = survival time): $(7.2 + 1.0 + 5.1 + 5.3 + 0.3)/5 = 3.78$. Also an **underestimate**, because IDs 16, 71, 55 actually survived *longer* than their follow-up times.

Acknowledging censoring. Writing censored times with a "+": 7.2^+, 1.0, 5.1^+, 5.3^+, 0.3. For ID 66 we know the patient survived exactly 1 year after MI; for ID 16 we only know that the patient survived *at least* 7.2 years. The data carry **incomplete (partial) information**.

Remark 1.3: Why both naive estimates are biased downward

Let T_i be the true survival time and C_i the follow-up (censoring) time, so we observe $Y_i = \min(T_i, C_i)$. Since $Y_i \leq T_i$ always, $\mathrm{E}(\bar{Y}) \leq \mathrm{E}(T)$ (approach 2). Approach 1 keeps only $\{T_i \leq C_i\}$, i.e. it conditions on the event time being *small enough* to be seen, which preferentially discards long survivors.

3 What Is Special About Survival Data?

- Classical mathematical statistics is built on i.i.d. data, e.g. n i.i.d. copies $T_1, \ldots, T_n$ of the survival time.
- The **incompleteness** of survival data distinguishes survival analysis from traditional methods.
- Formulating a survival problem requires three ingredients:

  (1) the **event of interest**;

  (2) the **time origin**;

  (3) the **time scale**.
- Incompleteness is realized as either **censoring** or **truncation**. Common types: right censoring (RC), left censoring (LC), interval censoring (IC), left truncation (LT), right truncation (RT).

4 Types of Censoring

4.1 Right censoring

Definition 1.4: Right censoring (RC)

For a right-censored observation, the exact event time is unknown; all we know is that the event happens *after* a certain time point: $T_i > C_i$.

Example 1.5: Time to death in a breast cancer trial

T_i = time to death from randomization (receipt of treatment); the time of randomization is the time origin. If the study ends 9 months after the patient's treatment date and the patient is still alive, T_i is right censored at 9 months: $T_i > 9$.

4.2 Left censoring

Definition 1.6: Left censoring (LC)

For a left-censored observation, all we know is that the event happened *prior to* a certain time point: $T_i < C_i$.

Example 1.7: Time to first use of marijuana

Question: "When did you first use marijuana?" Possible answers:

- the exact age (uncensored);

- "never used it" (right censored at the current age);
- "used it but cannot recall when" (left censored at the current age).

When both left and right censoring are present in the data we speak of **double censoring**.

4.3 Interval censoring

Definition 1.8: Interval censoring (IC)

For an interval-censored observation, all we know is that the event happened within an interval: $T_i \in (L_i, R_i]$.

Example 1.9: Time to tumour recurrence

After breast cancer surgery, MRI is performed every 5 months. If the tumour is absent at month 10 and detected at month 15, then $10 < T_i \leq 15$.

Remark 1.10

RC and LC are special cases of IC: right censoring at C is $T \in (C, \infty)$, left censoring at C is $T \in [0, C)$.

5 Types of Truncation

5.1 Left truncation

Definition 1.11: Left truncation (LT)

Only individuals who "survive" a sufficient time are **included** in the sample. If V_i is the entry (truncation) time, T_i is observed only if $T_i \geq V_i$, i.e. we observe the conditional distribution of T_i given $T_i \geq V_i$.

Example 1.12: Longevity of elderly residents of a retirement community

An individual must survive to a sufficient age to enter the retirement center; anyone who died earlier never enters. The data are left truncated. Ignoring LT leads to **length-biased sampling**: long lifetimes are over-represented.

5.2 Right truncation

Definition 1.13: Right truncation (RT)

Only individuals who have *experienced the event* by a specified time are included in the sample: T_i is observed only if $T_i < R_i$.

Example 1.14: Age at onset of diabetes

Diabetic patients are enrolled and asked retrospectively for their age of onset. With R_i = age at the time of the study and T_i = age of onset, T_i is observed conditionally on $T_i < R_i$: a subject must already have diabetes to be included.

Example 1.15: HIV incubation time (Lagakos et al., 1988)

Data on 258 adults and 37 children infected with HIV who developed AIDS by June 30, 1986. With X = age at HIV infection and R = age at developing AIDS, the incubation time $R - X$ is known only for those who were infected *and* developed AIDS within the study window. This is right truncation.

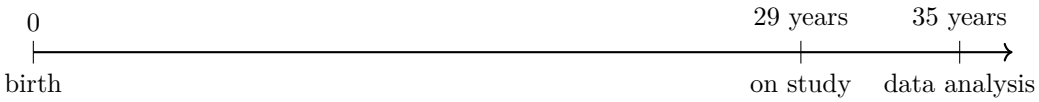
5.3 Censoring vs. truncation

Censoring	The event/subject *is detected*, but its value is not known completely: we only know it is less than and/or more than some number.
Truncation	A subject is detected *only if* its value is greater (or less) than some number; when detected, its value is completely known.

The key difference: a censored object is detectable, whereas a truncated object is not even detectable (it never enters the data set).

6 CVD Example: Time Scales, Truncation and Censoring Together

An observational study enrols subjects *without* heart disease to identify risk factors for cardiovascular disease (CVD); the event is CVD. Consider a subject who enters at age 29; data are analyzed when the subject would be 35.



Two time scales are possible depending on the origin:

- origin = birth: T_i^1 = age at CVD;
- origin = study entry: T_i^2 = years on study until CVD.

Both may be right censored (subject still CVD-free at the end of study). The first scale T_i^1 is usually more interesting.

Scenario	T_i^1 (age)	T_i^2 (time on study)
CVD at age 34	$T_i^1 = 34$ observed *given* $T_i^1 \geq 29$: **LT at 29**	$T_i^2 = 34 - 29 = 5$: no truncation, no censoring
No CVD by 35	$T_i^1 > 35$ given $T_i^1 \geq 29$: **LT at 29 and RC at 35**	$T_i^2 > 6$: **RC**, not LT
Enrolled at 30, already had CVD before 30	$T_i^1 < 30$: **LC at 30**	—

7 Definitions and Censoring Patterns

- **Failure time** (survival time, event time): the length of "time" to an event.
- **Time origin**: objectively defined and scientifically meaningful; ideally all subjects are comparable at the origin.
- **Time scale**: usually clock/calendar time (seconds, hours, weeks, months, years), but others are possible, e.g. QALY (quality-adjusted life years).

Definition 1.16: Equal or unequal censoring times

- **Type I censoring**: all subjects enter at the same real time and all censoring times are equal.
- **Generalized Type I censoring**: with staggered entry the censoring times differ, but each is known at entry.
- **Random censoring**: losses to follow-up (death from other causes, relocation, withdrawal of consent); censoring times are known only for those actually censored, and they differ.

It is crucial that the actual censoring *times* are recorded, not just the fact that a subject is censored.

8 Possible Simple (Naive) Methods

1. **Dichotomise at a pre-specified t_0.** Survivors to t_0 are "successes", deaths before t_0 "failures"; use binary-data methods (chi-square test, logistic regression).
   Disadvantages: difficulty with censoring before t_0; loss of efficiency from concentrating on a single point of the survival curve; arbitrary choice of t_0.
2. **Take logs and use the linear model.** Disadvantages: possible undue dependence on short failure times; the log-normal distribution may not be appropriate; still cannot handle censoring.

Both can be useful as a prelude to more complex analyses.

9 Summary

- Six types of incompleteness: RC, LC, IC, LT, RT (plus double censoring). LT and RC are the most common sampling schemes.
- **Censoring only**: T_i may not be observed directly; it is known to be $>$ or $<$ some value, or to lie in an interval.
- **Truncation only**: T_i is observed conditionally on $T_i >$ or $<$ some value.
- Mixtures of censoring and truncation are more complex.

10 Selected exercises

Motivation. The biases described above in words can be proved: under independence, discarding censored subjects shortens, left truncation lengthens and right truncation shortens the observed lifetimes, and the choice of time origin matters except for the exponential law.

Exercise 1.1 (Approach 1 is biased downward)

Let $T \geq 0$ with $\mathrm{E}(T) < \infty$ and let C be independent of T with $P(T \leq C) > 0$.
(a) Prove Chebyshev's association inequality: if φ is nondecreasing, g is nonincreasing, both are bounded or $\varphi(T), g(T), \varphi(T)g(T)$ are integrable, then $\mathrm{E}[\varphi(T)g(T)] \leq \mathrm{E}[\varphi(T)]\,\mathrm{E}[g(T)]$.
(b) Show that $\mathrm{E}(T \mid T \leq C) \leq \mathrm{E}(T)$.

Solution [Supplement].

(a) Let T' be an independent copy of T. For all s, t the numbers $\varphi(s) - \varphi(t)$ and $g(s) - g(t)$ have opposite signs (or one is 0), so

$$\{\varphi(T) - \varphi(T')\}\{g(T) - g(T')\} \leq 0.$$

Taking expectations and using that T, T' are i.i.d.,

$$2\mathrm{E}[\varphi(T)g(T)] - 2\mathrm{E}[\varphi(T)]\mathrm{E}[g(T)] \leq 0.$$

(b)

Step 1. Put $g(t) = P(C \geq t)$, which is nonincreasing with values in $[0, 1]$. By independence and Fubini, $P(T \leq C) = \mathrm{E}[g(T)]$ and $\mathrm{E}[T\, I(T \leq C)] = \mathrm{E}[T\, g(T)]$.

Step 2. Apply (a) with $\varphi(t) = t$ (integrable, since $\mathrm{E}T < \infty$ and $0 \leq g \leq 1$): $\mathrm{E}[Tg(T)] \leq \mathrm{E}(T)\mathrm{E}[g(T)]$.

Step 3. Dividing by $\mathrm{E}[g(T)] = P(T \leq C) > 0$,

$$\mathrm{E}(T \mid T \leq C) = \frac{\mathrm{E}[T\, I(T \leq C)]}{P(T \leq C)} \leq \mathrm{E}(T).$$

Approach 1 of the WHAS100 example estimates the left-hand side, so it is biased downward whatever the censoring distribution. ■

Exercise 1.2 (Truncation shifts the observed distribution)

Let $T \geq 0$ have survival function S.
(a) Let V be an entry time independent of T with $P(T \geq V) > 0$. Show that $P(T > t \mid T \geq V) \geq S(t)$ for every t, i.e. left-truncated lifetimes are stochastically larger than T.
(b) Let R be independent of T with $P(T < R) > 0$. Show that $P(T > t \mid T < R) \leq S(t)$ for every t.

Solution [Supplement].

(a) Let $g(s) = P(V \leq s)$, nondecreasing. By independence, $P(T \geq V) = \mathrm{E}[g(T)]$ and $P(T > t,\, T \geq V) = \mathrm{E}[I(T > t)g(T)]$. Both $s \mapsto I(s > t)$ and g are nondecreasing, so applying Exercise 1.1(a) to $\varphi = I(\cdot > t)$ and the nonincreasing $-g$ gives

$$\mathrm{E}[I(T > t)g(T)] \geq P(T > t)\,\mathrm{E}[g(T)].$$

Divide by $\mathrm{E}[g(T)] > 0$.

(b) Now $g(s) = P(R > s)$ is nonincreasing, $P(T < R) = \mathrm{E}[g(T)]$ and $P(T > t, T < R) = \mathrm{E}[I(T > t)g(T)]$. Exercise 1.1(a) with $\varphi = I(\cdot > t)$ gives $\mathrm{E}[I(T > t)g(T)] \le P(T > t)\mathrm{E}[g(T)]$; divide by $\mathrm{E}[g(T)] > 0$.

This is the precise form of length-biased sampling in the retirement-community example, and of the opposite bias in the diabetes example. ■

Exercise 1.3 (Changing the time origin)

Let T^1 (age at the event) be continuous with survival S_1 and hazard h_1, and let $a \ge 0$ with $S_1(a) > 0$. A subject enters the study at age a and is followed on the study scale $T^2 = T^1 - a$, conditionally on $T^1 \ge a$.
(a) Show that, given $T^1 \ge a$, T^2 has survival $S_1(a+u)/S_1(a)$ and hazard $h_1(a+u)$, $u \ge 0$.
(b) Assume $S_1 > 0$ on $[0, \infty)$. Show that the conditional law of T^2 equals the law of T^1 for every $a \ge 0$ if and only if T^1 is exponential.

Solution [Supplement].

(a) For $u \ge 0$, $P(T^2 > u \mid T^1 \ge a) = P(T^1 > a+u)/P(T^1 \ge a) = S_1(a+u)/S_1(a)$, using continuity. Taking $-\frac{\mathrm{d}}{\mathrm{d}u} \log$ gives the hazard $-\frac{\mathrm{d}}{\mathrm{d}u} \log S_1(a+u) = h_1(a+u)$.

(b)

Step 1. If $T^1 \sim \mathrm{Exp}(\lambda)$, then $S_1(a+u)/S_1(a) = e^{-\lambda u} = S_1(u)$.

Step 2. Conversely, suppose $S_1(a+u) = S_1(a)S_1(u)$ for all $a, u \ge 0$. Then $\psi = -\log S_1$ (finite, since $S_1 > 0$) satisfies $\psi(a+u) = \psi(a) + \psi(u)$, $\psi \ge 0$ nondecreasing.

Step 3. Additivity gives $\psi(q) = q\,\psi(1)$ for rational $q \ge 0$. For real $x \ge 0$ and rationals $q_1 \le x \le q_2$, monotonicity gives $q_1\psi(1) \le \psi(x) \le q_2\psi(1)$; letting $q_1, q_2 \to x$ yields $\psi(x) = \lambda x$ with $\lambda = \psi(1)$. Since $S_1(\infty) = 0$, $\lambda > 0$, so $T^1 \sim \mathrm{Exp}(\lambda)$.

Hence, outside the exponential model, analyzing the CVD study on the age scale (with left truncation) and on the study scale (without) answers different questions. ■